

Lecture 8: Infinite-Horizon Optimization and Dynamic Programming

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- ▶ Brief introduction to infinite-horizon optimization in discrete time under certainty
- ▶ Main purpose: introduce the theory and techniques of dynamic programming

The material is presented in four parts:

- 1 Introduces the problem and provides theoretical results necessary for applications of stationary dynamic programming techniques
- 2 Additional mathematical tools
- 3 Applications

- 1 Problem Description
- 2 Stationary Dynamic Programming
- 3 Stationary Dynamic Programming Results
- 4 The Contraction Mapping Theorem
- 5 Proofs of Main Theorems
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 - Euler Equation
 - Dynamic Programming vs Sequence Problem

The canonical discrete-time infinite-horizon optimization program:

$$\begin{aligned} & \sup_{\{x_t, y_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \tilde{U}(t, x_t, y_t) \\ \text{s.t. } & y_t \in \tilde{G}(t, x_t) \text{ for all } t \geq 0 \\ & x_{t+1} = \tilde{f}(t, x_t, y_t) \text{ for all } t \geq 0 \\ & x_0 \text{ given} \end{aligned}$$

- ▶ $\beta \in [0, 1)$ is the discount factor
- ▶ $x_t \in X \subset \mathbb{R}^{K_x}$ and $y_t \in Y \subset \mathbb{R}^{K_y}$ for some $K_x, K_y \geq 1$
- ▶ x_t : state variables (state vector)
- ▶ y_t : control variables (control vector)
- ▶ $\tilde{U} : \mathbb{Z}_+ \times X \times Y \rightarrow \mathbb{R}$ is the instantaneous payoff function
- ▶ $\tilde{G}(t, x)$ is a set-valued mapping (correspondence): $\tilde{G} : \mathbb{Z}_+ \times X \rightrightarrows Y$

- ▶ First constraint: $y_t \in \tilde{G}(t, x_t)$ specifies what values of control vector y_t are allowed, given state x_t at time t
- ▶ Evolution equation: $\tilde{f} : \mathbb{Z}_+ \times X \times Y \rightarrow X$ specifies evolution of state vector as function of last period's state and control vectors
- ▶ This formulation highlights distinction between state and control variables
- ▶ Often more convenient to work with transformation to eliminate y_t

$$V^*(0, x_0) = \sup_{\{x_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(t, x_t, x_{t+1})$$

s.t. $x_{t+1} \in G(t, x_t)$ for all $t \geq 0$
 x_0 given

- ▶ $x_t \in X \subset \mathbb{R}^K$ (now $K = K_x$)
- ▶ x_t corresponds to state vector, x_{t+1} plays role of control vector at time t
- ▶ $U : \mathbb{Z}_+ \times X \times X \rightarrow \mathbb{R}$ is instantaneous payoff function
- ▶ $G : \mathbb{Z}_+ \times X \rightrightarrows X$ specifies constraint correspondence

- ▶ $V^* : \mathbb{Z}_+ \times X \rightarrow \mathbb{R}$ is the **value function**, which specifies supremum (highest possible value) that objective function can reach starting with some x_t at time t
- ▶ When maximal value is attained by sequence $\{x_{t+1}^*\}_{t=0}^\infty \in X^\infty$, we refer to this as a **solution** or an **optimal plan**
- ▶ Objective: characterize optimal plan $\{x_{t+1}^*\}_{t=0}^\infty$ and value function $V^*(0, x_0)$
- ▶ Here, X^∞ is countable infinite product of set X . If X is bounded, then $X^\infty \subset \ell^\infty$, where ℓ^∞ is the vector space of infinite sequences that are bounded with the sup norm $\|\cdot\|_\infty$ (or simply $\|\cdot\|$)

Example 6.1: Optimal Growth Problem

Consider an optimal growth problem:

$$\begin{aligned} & \max_{\{k_t, c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t. } & k_{t+1} = f(k_t) + (1 - \delta)k_t - c_t \\ & k_t \geq 0, \text{ given } k_0 > 0, \text{ and } u : \mathbb{R}_+ \rightarrow \mathbb{R} \end{aligned}$$

Let $x_t = k_t$ and $x_{t+1} = k_{t+1}$. The objective function can be written as:

$$\max_{\{k_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(f(k_t) - k_{t+1} + (1 - \delta)k_t)$$

subject to $k_t \geq 0$. This is special case of Problem 6.1 with:

- ▶ $U(t, k_t, k_{t+1}) = u(f(k_t) - k_{t+1} + (1 - \delta)k_t)$
- ▶ Constraint correspondence $G(t, k_t) = [0, f(k_t) + (1 - \delta)k_t]$ (since $c_t \geq 0$)

- ▶ Notable feature emphasized by Example 6.1: U and G do not explicitly depend on time
- ▶ This feature is fairly common in economics and many interesting problems can be formulated in **stationary form**
- ▶ Stationary problem involves:
 - ▶ Objective function that is a discounted sum
 - ▶ U and G functions that do not explicitly depend on time

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Consider the stationary form of Problem 6.1.

Problem 6.2:

$$V^*(x_0) = \sup_{\{x_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(x_t, x_{t+1})$$

s.t. $x_{t+1} \in G(x_t)$ for all $t \geq 0$
 x_0 given

- ▶ Constraint correspondence: $G : X \rightrightarrows X$
- ▶ Instantaneous payoff functions: $U : X \times X \rightarrow \mathbb{R}$
- ▶ Value function without time argument: $V^*(x_0)$

- ▶ Problem 6.2 corresponds to the **sequence problem**: involves choosing infinite sequence $\{x_t\}_{t=0}^{\infty} \in X^{\infty}$
- ▶ Sequence problems sometimes have nice features, but solutions often difficult to characterize analytically and numerically
- ▶ Basic idea of dynamic programming: turn sequence problem into **functional equation**
- ▶ Transform problem into one of finding a function rather than a sequence

Problem 6.3:

$$V(x) = \sup_{y \in G(x)} \{U(x, y) + \beta V(y)\}, \quad \text{for all } x \in X$$

where $V : X \rightarrow \mathbb{R}$.

- ▶ This functional equation is also called the **“Bellman equation”** after Richard Bellman
- ▶ Instead of explicitly choosing a sequence $\{x_t\}_{t=0}^{\infty}$, we choose a **policy** $y \in G(x) \subset X$ for given $x \in X$
- ▶ Since $U(\cdot, \cdot)$ does not depend on time, no reason for policy to be time-dependent either
- ▶ Use V for function defined in Problem 6.3 and V^* for function defined in Problem 6.2

$$V(x) = \sup_{y \in G(x)} \{U(x, y) + \beta V(y)\}, \quad \text{for all } x \in X$$

- ▶ This is a “**recursive formulation**”: function $V(x)$ appears on both left- and right-hand sides, defined recursively
- ▶ In many cases, solution to Problem 6.3 is simpler to characterize analytically than the corresponding solution to sequence problem
- ▶ Question: when are the two problems equivalent?

Suppose Problem 6.2 has maximum starting at x_0 attained by optimal sequence $\{x_t^*\}_{t=0}^{\infty}$ with $x_0^* = x_0$. Under weak technical conditions:

$$\begin{aligned} V^*(x_0) &= \sum_{t=0}^{\infty} \beta^t U(x_t^*, x_{t+1}^*) \\ &= U(x_0, x_1^*) + \beta \sum_{s=0}^{\infty} \beta^s U(x_{s+1}^*, x_{s+2}^*) \\ &= U(x_0, x_1^*) + \beta V^*(x_1^*) \end{aligned}$$

This encapsulates the basic idea of dynamic programming: **Principle of Optimality**, which states that optimal plan can be broken into two parts:

- 1 What is optimal today
- 2 The optimal continuation path

$$V(x) = \sup_{y \in G(x)} \{U(x, y) + \beta V(y)\}, \quad \text{for all } x \in X$$

- ▶ Solution can be represented by a time invariant **policy function** (or policy mapping) $\pi : X \rightarrow X$ that determines which value of x_{t+1} to choose for given state variable x_t
- ▶ Two complications:
 - ① Control reaching optimal value may not exist (reason for “sup” notation)
 - ② Solution may involve policy correspondence $\Pi : X \rightrightarrows X$ (more than one maximizer)
- ▶ Ignore these complications for now

Once value function V is determined, policy function is straightforward to characterize.
If the optimal policy is given by $\pi(x)$, then:

$$V(x) = U(x, \pi(x)) + \beta V(\pi(x)) \quad \text{for all } x \in X$$

- ▶ This equation follows from the fact that $\pi(x)$ is the optimal policy
- ▶ When $y = \pi(x)$, the right-hand side of the Bellman equation reaches maximal value $V(x)$
- ▶ This is one way of determining the policy function

- ▶ There are powerful tools that establish existence of solution and some of its properties
- ▶ Next section: present results on relationship between solution to the sequence problem (Problem 6.2) and the recursive formulation (Problem 6.3)
- ▶ Establish results on concavity, monotonicity, and differentiability of the value function

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Main purpose:

- ▶ Ensure sequence $\{x_t^*\}_{t=0}^\infty \in X^\infty$ that attains supremum in Problem 6.2 satisfies recursive equation of dynamic programming

$$V(x_t^*) = U(x_t^*, x_{t+1}^*) + \beta V(x_{t+1}^*), \quad \text{for all } t = 0, 1, \dots$$

- ▶ The solution to the Bellman equation is also a solution to Problem 6.2
- ▶ In other words, establish equivalence results between solutions to Problems 6.2 and 6.3

Define set of feasible sequences or plans starting with initial value x_t :

$$\Phi(x_t) = \{\{x_s\}_{s=t}^{\infty} : x_{s+1} \in G(x_s) \text{ for } s = t, t+1, \dots\}$$

- ▶ Intuitively, $\Phi(x_t)$ is the set of feasible choices of vectors starting from x_t
- ▶ Denote typical element of $\Phi(x_0)$ by $\mathbf{x} = (x_0, x_1, \dots) \in \Phi(x_0)$
- ▶ Want sequence to satisfy:

$$V(x_t^*) = U(x_t^*, x_{t+1}^*) + \beta V(x_{t+1}^*) \text{ for all } t = 0, 1, \dots$$

Assumption 6.1

$G(x)$ is nonempty for all $x \in X$; and for all $x_0 \in X$ and $\mathbf{x} \in \Phi(x_0)$,

$$\lim_{n \rightarrow \infty} \sum_{t=0}^n \beta^t U(x_t, x_{t+1}) \text{ exists and is finite}$$

- ▶ Assumption stronger than necessary for theory: for much of dynamic programming theory, it is sufficient that the limit exists
- ▶ But in economic applications, we are not interested in optimization problems where agents achieve infinite value

Theorem 6.1 (Equivalence of Values)

Suppose Assumption 6.1 holds. Then for any $x \in X$, any solution $V^*(x)$ to Problem 6.2 is also a solution to Problem 6.3. Moreover, any solution $V(x)$ to Problem 6.3 is also a solution to Problem 6.2, so that $V^*(x) = V(x)$ for all $x \in X$.

- ▶ Under Assumption 6.1, both sequence problem and recursive formulation achieve the same value
- ▶ Fundamental result that justifies the dynamic programming approach

Theorem 6.2 (Principle of Optimality)

Suppose Assumption 6.1 holds. Let $\mathbf{x}^* \in \Phi(x_0)$ be a feasible plan that attains $V^*(x_0)$ in Problem 6.2. Then

$$V^*(x_t^*) = U(x_t^*, x_{t+1}^*) + \beta V^*(x_{t+1}^*) \quad (6.3)$$

for $t = 0, 1, \dots$, with $x_0^* = x_0$.

Moreover, if any $\mathbf{x}^* \in \Phi(x_0)$ satisfies equation (6.3), then it attains the optimal value in Problem 6.2.

- ▶ If any feasible plan satisfies the Bellman equation, then it attains optimal value
- ▶ We can go from solution of the Bellman equation to solution of the sequence problem and vice versa

Example 6.4: Consider an optimal growth model with log preferences, Cobb-Douglas technology, and full depreciation:

$$\begin{aligned} \max_{\{k_t, c_t\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \log c_t \\ \text{s.t.} \quad & k_{t+1} = k_t^\alpha - c_t \\ & k_0 > 0 \end{aligned}$$

where $\beta \in (0, 1)$, k is capital-labor ratio, and the resource constraint follows from the production function $K^\alpha L^{1-\alpha}$ in per capita terms.

We solve the Bellman equation

$$V(x) = \max_{y \geq 0} \{ \log(x^\alpha - y) + \beta V(y) \}.$$

Standard approach for log-Cobb-Douglas:

- Conjecture that the value function is logarithmic:

$$V(x) = A + B \log x.$$

- Use the Bellman equation to solve for unknown constants (A, B) .
- Obtain closed-form policy and value functions.

- Guess:

$$V(x) = A + B \log x.$$

- Bellman equation:

$$A + B \log x = \max_y \{ \log(x^\alpha - y) + \beta [A + B \log y] \}.$$

- FOC w.r.t. y :

$$-\frac{1}{x^\alpha - y} + \frac{\beta B}{y} = 0 \implies y = \frac{\beta B}{1 + \beta B} x^\alpha.$$

Using $B = \frac{\alpha}{1 - \alpha\beta}$ gives $y = \alpha\beta x^\alpha$.

- Optimal next-period capital:

$$k_{t+1} = y^*(x) = \alpha\beta x^\alpha.$$

- ▶ Substitute $y^*(x) = \alpha\beta x^\alpha$ into Bellman equation:

$$\begin{aligned} A + B \log x &= \log((1 - \alpha\beta)x^\alpha) + \beta [A + B \log(\alpha\beta x^\alpha)] \\ &= \log(1 - \alpha\beta) + \alpha \log x + \beta A + \beta B (\log(\alpha\beta) + \alpha \log x). \end{aligned}$$

- ▶ Match coefficients on $\log x$:

$$B = \alpha + \beta B \alpha \implies B = \frac{\alpha}{1 - \alpha\beta}$$

- ▶ Match constant terms:

$$A = \log(1 - \alpha\beta) + \beta A + \beta B \log(\alpha\beta).$$

Solve:

$$A = \frac{1}{1 - \beta} \left[\log(1 - \alpha\beta) + \frac{\alpha\beta}{1 - \alpha\beta} \log(\alpha\beta) \right].$$

Assumption 6.2

X is a compact subset of $\mathbb{R}^K$, G is nonempty-valued, compact-valued, and continuous. Moreover, $U : X_G \rightarrow \mathbb{R}$ is continuous, where $X_G = \{(x, y) \in X \times X : y \in G(x)\}$.

- ▶ Most restrictive: the state space is compact
- ▶ Since X is compact and $G(x)$ is compact-valued, X_G is also compact
- ▶ Continuous function on compact domain: U is bounded

Theorem 6.3 (Existence of Solutions)

Suppose that Assumptions 6.1 and 6.2 hold. Then there exists a unique continuous and bounded function $V : X \rightarrow \mathbb{R}$ that satisfies the Bellman equation. Moreover, for any $x_0 \in X$, an optimal plan $\mathbf{x}^* \in \Phi(x_0)$ exists.

Two major results:

- 1 **Existence and uniqueness** of the value function
- 2 **Existence** of an optimal plan

Combined with Theorem 6.1: optimal policy function achieving supremum V^* in Problem 6.2 exists and V^* is continuous and bounded.

- ▶ Optimal plan in Problem 6.2 (or 6.3) may not be unique, even though the value function is unique: for example, when two alternative feasible sequences achieve same maximal value
- ▶ As in static optimization problems, non-uniqueness because of the lack of strict concavity of objective function
- ▶ When we include a concavity assumption, uniqueness of optimal plan is guaranteed

Assumption 6.3

U is concave; that is, for any $\alpha \in (0, 1)$ and any $(x, y), (x', y') \in X_G$:

$$U(\alpha x + (1 - \alpha)x', \alpha y + (1 - \alpha)y') \geq \alpha U(x, y) + (1 - \alpha)U(x', y')$$

and the inequality is strict if $x \neq x'$.

In addition, G is convex in the sense that for any $\alpha \in [0, 1]$, and $x, x', y, y' \in X$ such that $y \in G(x)$ and $y' \in G(x')$:

$$\alpha y + (1 - \alpha)y' \in G(\alpha x + (1 - \alpha)x').$$

Theorem 6.4 (Concavity of the Value Function)

Suppose that Assumptions 6.1, 6.2, and 6.3 hold. Then the unique $V : X \rightarrow \mathbb{R}$ that satisfies the Bellman equation is strictly concave.

- ▶ If Assumption 6.3 relaxed so that U is concave (without additional requirement for $x \neq x'$), then a weaker version applies and implies V is concave
- ▶ Important for uniqueness of optimal plans

Corollary 6.1

Suppose Assumptions 6.1, 6.2, and 6.3 hold. Then there exists a unique optimal plan $x^* \in \Phi(x_0)$ for any $x_0 \in X$. Moreover, the optimal plan can be expressed as:

$$x_{t+1}^* = \pi(x_t^*)$$

where $\pi : X \rightarrow X$ is a continuous policy function.

- ▶ The policy correspondence becomes a function, not just a correspondence.
- ▶ The policy function is continuous.

Assumption 6.4

For each $y \in X$, $U(\cdot, y)$ is strictly increasing in each of its first K arguments, and G is monotone in the sense that $x \leq x'$ implies $G(x) \subset G(x')$.

- ▶ Ensures payoff function increasing in state variables
- ▶ Larger values of state variables attractive from viewpoint of relaxing constraints
- ▶ Natural in economic applications where “more is better”

Theorem 6.5 (Monotonicity)

Suppose Assumptions 6.1, 6.2, and 6.4 hold, and let $V : X \rightarrow \mathbb{R}$ be the unique solution to the Bellman equation. Then V is strictly increasing in all of its arguments.

- ▶ Consistent with monotonic preferences
- ▶ Useful for comparative statics analysis
- ▶ In growth models, higher capital stock always increases value function

Assumption 6.5

U is continuously differentiable on the interior of its domain X_G .

- ▶ Common in most economic models
- ▶ Enables us to work with first-order necessary conditions

Theorem 6.6 (Differentiability)

Suppose Assumptions 6.1, 6.2, 6.3, and 6.5 hold. Let $\pi(\cdot)$ be the policy function from Corollary 6.1, and assume $x \in \text{Int } X$ and $\pi(x) \in \text{Int } G(x)$. Then $V(\cdot)$ is differentiable at x , with gradient:

$$DV(x) = D_x U(x, \pi(x))$$

- ▶ Allows us to use differential calculus on the Bellman equation
- ▶ Enables derivation of Euler equations

Given these assumptions, the following results are established:

- ① **Theorem 6.1:** Equivalence of Values
- ② **Theorem 6.2:** Principle of Optimality
- ③ **Theorem 6.3:** Existence of Solutions
- ④ **Theorem 6.4:** Concavity of Value Function
- ⑤ **Theorem 6.5:** Monotonicity of Value Function
- ⑥ **Theorem 6.6:** Differentiability of Value Function and the Envelope Theorem

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Metric Space: (S, d) where S is a space and d is a distance metric

Operators: Mappings $T : S \rightarrow S$

Definition 6.1 (Contraction Mapping)

Let (S, d) be a metric space and $T : S \rightarrow S$ be an operator. If for some $\beta \in (0, 1)$:

$$d(Tz_1, Tz_2) \leq \beta d(z_1, z_2) \text{ for all } z_1, z_2 \in S$$

then T is a contraction mapping (with modulus β).

Intuition: Contraction mappings bring elements uniformly closer together.

Example 6.2: On interval $S = [a, b]$ with usual metric $d(z_1, z_2) = |z_1 - z_2|$: T is a contraction if:

$$\frac{|Tz_1 - Tz_2|}{|z_1 - z_2|} \leq \beta < 1 \text{ for all } z_1 \neq z_2$$

This means T has slope less than 1 everywhere.

The Contraction Mapping Theorem

Definition 6.2 (Fixed Point)

A fixed point of T is any element $\hat{z} \in S$ satisfying $T\hat{z} = \hat{z}$.

Theorem 6.7 (Contraction Mapping Theorem)

Let (S, d) be a complete metric space and suppose $T : S \rightarrow S$ is a contraction. Then T has a unique fixed point $\hat{z}$; that is, there exists a unique $\hat{z} \in S$ such that:

$$T\hat{z} = \hat{z}.$$

Moreover, $T^n z \rightarrow \hat{z}$ as $n \rightarrow \infty$ for any $z \in S$.

- ▶ Simple conditions that guarantee existence, uniqueness, and convergence
- ▶ Applies to infinite-dimensional function spaces
- ▶ Foundation of dynamic programming theory

Theorem 6.8

Let (S, d) be complete and $T : S \rightarrow S$ be a contraction with fixed point $\hat{z}$.

- ① If S' is a closed subset of S and $T(S') \subseteq S'$, then $\hat{z} \in S'$
- ② If $T(S') \subseteq S'' \subseteq S'$, then $\hat{z} \in S''$

- ▶ If we start in a closed set that maps to itself, fixed point is in that set
- ▶ Enables proving properties like strict concavity or monotonicity

Checking contraction property directly is often difficult. Blackwell provides easier conditions:

Theorem 6.9 (Blackwell's Conditions)

Let $X \subseteq \mathbb{R}^K$ and $B(X)$ be the space of bounded functions $f : X \rightarrow \mathbb{R}$ equipped with the sup norm. If $T : B(X) \rightarrow B(X)$ satisfies:

- ① Monotonicity: For any $f, g \in B(X)$, $f(x) \leq g(x)$ for all x implies $(Tf)(x) \leq (Tg)(x)$ for all $x \in X$; and
- ② Discounting: $\exists \beta \in (0, 1)$ such that

$$[T(f + c)](x) \leq (Tf)(x) + \beta c$$

for all $f \in B(X)$, $c \geq 0$, and $x \in X$.

Then T is a contraction with modulus β on $B(X)$.

For the Bellman operator:

$$TV(x) = \max_{y \in G(x)} \{U(x, y) + \beta V(y)\}$$

- ▶ Checking Monotonicity: If $V_1(x) \leq V_2(x)$ for all x , then:

$$TV_1(x) = \max_{y \in G(x)} \{U(x, y) + \beta V_1(y)\} \leq \max_{y \in G(x)} \{U(x, y) + \beta V_2(y)\} = TV_2(x)$$

- ▶ Checking Discounting:

$$\begin{aligned} T(V + c)(x) &= \max_{y \in G(x)} \{U(x, y) + \beta(V(y) + c)\} \\ &= \max_{y \in G(x)} \{U(x, y) + \beta V(y)\} + \beta c \\ &= TV(x) + \beta c \end{aligned}$$

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For a feasible infinite sequence $x = (x_0, x_1, \dots) \in \Phi(x_0)$, define

$$\bar{U}(x) \equiv \sum_{t=0}^{\infty} \beta^t U(x_t, x_{t+1})$$

Lemma 6.1

Suppose Assumption 6.1 holds. For any $x_0 \in X$ and $x \in \Phi(x_0)$:

$$\bar{U}(x) = U(x_0, x_1) + \beta \bar{U}(x')$$

where $x' = (x_1, x_2, \dots)$.

Goal: Show $V^*(x) = V(x)$ where:

- ▶ $V^*(x) = \sup_{z \in \Phi(x)} \bar{U}(z)$ (Problem 6.2)
- ▶ $V(x) = \sup_{y \in G(x)} \{U(x, y) + \beta V(y)\}$ (Problem 6.3)

Idea: Use Lemma 6.1 and the definition of supremum. (Read pp. 195–197 of the textbook.)

- ▶ Let x^* attain $V^*(x_0)$ in Problem 6.2. Our goal is to show that

$$V^*(x_t^*) = U(x_t^*, x_{t+1}^*) + \beta V^*(x_{t+1}^*)$$

Proof idea: show that $(x_t^*, x_{t+1}^*, \dots)$ is optimal starting from x_t^* using Lemma 6.1 and induction

- ▶ Suppose x^* satisfies the Bellman equation. We want to show that it is optimal.

Proof idea: iterating on the Bellman equation.

Version 1 (Sequence problem under compact X):

- ▶ Basic idea: show that the objective function is continuous on a compact set, and therefore attains a maximum
- ▶ Treat the objective function as continuous on X^∞
- ▶ Show that the constraint set is a compact subset of X^∞

Version 2 (Contraction Mapping):

- ▶ Define Bellman operator T on space $C(X)$ of continuous functions
- ▶ Show T satisfies Blackwell's conditions
- ▶ Apply Contraction Mapping Theorem: unique fixed point exists
- ▶ Fixed point is solution to recursive problem

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- ▶ We show how dynamic programming can be applied to a range of problems
- ▶ **Main Result:** Theorem 6.10 shows how dynamic first-order conditions (Euler equations) together with the transversality condition are sufficient to characterize solutions

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Consider the functional equation corresponding to Problem 6.3:

$$V(x) = \max_{y \in G(x)} \{U(x, y) + \beta V(y)\} \quad \text{for all } x \in X$$

Throughout, we assume Assumptions 6.1–6.5 hold.

- ▶ From Theorem 6.4: maximization problem is strictly concave
- ▶ From Theorem 6.6: maximand is differentiable
- ▶ For interior solutions $y \in \text{Int } G(x)$: first-order conditions are necessary and sufficient

For optimal solutions, we can characterize them by the **Euler equations**:

$$D_y U(x, y^*) + \beta D V(y^*) = 0$$

- Use the Envelope Theorem for dynamic programming by differentiating the functional equation with respect to state vector x :

$$D V(x) = D_x U(x, y^*)$$

- Substituting it into the Euler equation gives

$$D_y U(x, \pi(x)) + \beta D_x U(\pi(x), \pi(\pi(x))) = 0$$

This is a **functional equation** in the unknown function $\pi(\cdot)$ and characterizes the optimal policy function.

When both x and y are real numbers, the Euler equation becomes:

$$\frac{\partial U(x, y^*)}{\partial y} + \beta V'(y^*) = 0$$

Sum of marginal gain today from increasing y plus discounted marginal gain from increasing y on future returns must equal zero

The one-dimensional Envelope Condition gives:

$$V'(x) = \frac{\partial U(x, y^*)}{\partial x}$$

Combining with the Euler equation:

$$\frac{\partial U(x, \pi(x))}{\partial y} + \beta \frac{\partial U(\pi(x), \pi(\pi(x)))}{\partial x} = 0$$

With time arguments explicitly:

$$\frac{\partial U(x_t, x_{t+1}^*)}{\partial y} + \beta \frac{\partial U(x_{t+1}^*, x_{t+2}^*)}{\partial x} = 0$$

- ▶ Euler equation is **not sufficient** for optimality. We also need the transversality condition.
- ▶ The transversality condition ensures that there are no beneficial simultaneous changes in an infinite number of choice variables.

- ▶ **General case:**

$$\lim_{t \rightarrow \infty} \beta^t D_x U(x_t^*, x_{t+1}^*) \cdot x_t^* = 0$$

- ▶ **One-dimensional case:**

$$\lim_{t \rightarrow \infty} \beta^t \frac{\partial U(x_t^*, x_{t+1}^*)}{\partial x} \cdot x_t^* = 0$$

- ▶ The transversality condition requires that the marginal return from state variable x times the value of this state variable does not increase asymptotically at rate faster than or equal to $1/\beta$.

Theorem 6.10 (Euler Equations and Transversality Condition)

Let $X \subset \mathbb{R}_+^K$ and suppose Assumptions 6.1–6.5 hold. Then a sequence $\{x_t^*\}_{t=0}^\infty$ such that $x_{t+1}^* \in \text{Int } G(x_t^*)$, $t = 0, 1, \dots$, is optimal for Problem 6.2 given x_0 if and only if it satisfies:

$$\begin{aligned} D_y U(x_t^*, x_{t+1}^*) + \beta D_x U(x_{t+1}^*, x_{t+2}^*) &= 0 \\ \lim_{t \rightarrow \infty} \beta^t D_x U(x_t^*, x_{t+1}^*) \cdot x_t^* &= 0 \end{aligned}$$

The Euler equations together with the transversality condition are **necessary and sufficient**, under those assumptions.

Example 6.4: Consider an optimal growth model with log preferences, Cobb-Douglas technology, and full depreciation:

$$\begin{aligned} \max_{\{k_t, c_t\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \log c_t \\ \text{s.t.} \quad & k_{t+1} = k_t^\alpha - c_t \\ & k_0 > 0 \end{aligned}$$

where $\beta \in (0, 1)$, k is capital-labor ratio, and the resource constraint follows from the production function $K^\alpha L^{1-\alpha}$ in per capita terms.

Write down the Bellman equation

$$V(x) = \max_{y \geq 0} \{ \log(x^\alpha - y) + \beta V(y) \}$$

where x is today's capital stock and y is tomorrow's capital stock.

- ▶ Previously, we used guess and verify to solve the value function.
- ▶ Show that the problem satisfies Assumptions 6.1–6.5, so Theorems 6.1–6.6 apply. Then we can also use the Euler equation.

- ▶ Since $V(\cdot)$ is differentiable, the first-order condition is

$$\frac{1}{x^\alpha - y} = \beta V'(y)$$

- ▶ Envelope Condition:

$$V'(x) = \frac{\alpha x^{\alpha-1}}{x^\alpha - y}$$

- ▶ Combining using $y = \pi(x)$:

$$\frac{1}{x^\alpha - \pi(x)} = \beta \frac{\alpha \pi(x)^{\alpha-1}}{\pi(x)^\alpha - \pi(\pi(x))}$$

This is a functional equation in a single function $\pi(x)$.

- ▶ There are no straightforward ways to solve functional equations, but guess-and-verify often works.
- ▶ Let us conjecture that $\pi(x) = ax^\alpha$
- ▶ Substituting into the Euler equation:

$$\frac{1}{x^\alpha - ax^\alpha} = \beta \frac{\alpha a^{\alpha-1} x^{\alpha(\alpha-1)}}{a^\alpha x^{\alpha^2} - a^{1+\alpha} x^{\alpha^2}} = \frac{\beta}{a} \frac{\alpha}{x^\alpha - ax^\alpha}$$

which implies $a = \beta\alpha$ satisfies the equation.

- ▶ The policy function is: $\pi(x) = \beta\alpha x^\alpha$
- ▶ Law of motion: $k_{t+1} = \beta\alpha k_t^\alpha$
- ▶ Optimal consumption: $c_t = (1 - \beta\alpha)k_t^\alpha$
- ▶ Capital-labor ratio k_t converges to steady state k^* , which ensures transversality condition
- ▶ By Corollary 6.1 and Theorem 6.10: $\pi(x) = \beta\alpha x^\alpha$ is the unique policy function

Example 6.5: Consider an infinitely-lived consumer who solves

$$\max_{\{c_t, a_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to: $a_{t+1} = (1 + r)a_t + w - c_t, \quad c_t \geq 0$

- ▶ $u(c)$ is strictly increasing, continuously differentiable, strictly concave
- ▶ Without further constraints, this is not well-defined and allows consumer to build unlimited debt ($\lim_{t \rightarrow \infty} a_t = -\infty$)—"Ponzi games".

Three approaches to prevent unlimited borrowing:

- ① **No-Ponzi condition:** Rule out such schemes directly
- ② **No borrowing:** $a_{t+1} \geq 0$ for all t (too restrictive)
- ③ **Natural debt limit:** Maximum repayable debt $a_{t+1} \geq \underline{a} \equiv -w/r$

- ▶ **Challenge:** Assets a don't necessarily belong to a compact set.
- ▶ **Solution:** Choose upper bound $\bar{a}$ and restrict $a \in [\underline{a}, \bar{a}]$.
 - ① Solve problem on a compact set
 - ② Verify that the solution indeed lies in the state space
- ▶ **Natural choice:** $\bar{a} \equiv a_0 + w/r < \infty$

- ▶ State variable: a_t
- ▶ Consumption: $c_t = (1 + r)a_t + w - a_{t+1} \geq 0$
- ▶ The Bellman equation

$$V(a) = \max_{a' \in [\underline{a}, \bar{a}], a' \leq (1+r)a + w} \{u((1+r)a + w - a') + \beta V(a')\}$$

Verify that the assumptions are satisfied.

- Using one-dimensional Euler equation and the Envelope Condition gives

$$u'(c) = \beta(1+r)u'(c')$$

- Since $u'(\cdot)$ exists and is strictly decreasing (because u is continuously differentiable and strictly concave), we get the following simple rule:

if $r = \beta^{-1} - 1$, $c = c'$ (constant consumption)

if $r > \beta^{-1} - 1$, $c < c'$ (increasing consumption)

if $r < \beta^{-1} - 1$, $c > c'$ (decreasing consumption)

- 1 Problem Description
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 - Euler Equation
 - Dynamic Programming vs Sequence Problem

Consider a dynamic optimization problem with finite horizon T :

$$\begin{aligned} \max_{\{x_t\}_{t=1}^{T+1}} \quad & \sum_{t=0}^T \beta^t U(x_t, x_{t+1}) \\ \text{s.t.} \quad & x_{t+1} \geq 0, \quad x_0 \text{ given} \end{aligned}$$

- ▶ When $0 \leq t \leq T-1$, FOCs from finite-dimensional optimization assuming interior solution:

$$\frac{\partial U(x_t^*, x_{t+1}^*)}{\partial y} + \beta \frac{\partial U(x_{t+1}^*, x_{t+2}^*)}{\partial x} = 0$$

- ▶ When $t = T$, the complementary slackness condition implies:

$$x_{T+1}^* \geq 0, \text{ and } \beta^T \frac{\partial U(x_T^*, x_{T+1}^*)}{\partial y} x_{T+1}^* = 0$$

Example 6.6: In the optimal growth problem,

$$U(x_t, x_{t+1}) = u(f(x_t) + (1 - \delta)x_t - x_{t+1})$$

Suppose that the world ends at date T . Then at the last date T :

$$\frac{\partial U(x_T^*, x_{T+1}^*)}{\partial y} = -u'(c_T^*) < 0$$

From the boundary condition, an optimal path must have $k_{T+1}^* = x_{T+1}^* = 0$.

No capital should be left at end of world. Any leftover resources should be consumed.

- ▶ The transversality condition can be derived heuristically by taking the limit of the boundary condition

$$\lim_{T \rightarrow \infty} \beta^T \frac{\partial U(x_T^*, x_{T+1}^*)}{\partial y} x_{T+1}^* = 0$$

- ▶ The Euler equation implies

$$\frac{\partial U(x_T^*, x_{T+1}^*)}{\partial y} + \beta \frac{\partial U(x_{T+1}^*, x_{T+2}^*)}{\partial x} = 0$$

- ▶ Substituting and changing timing:

$$\lim_{T \rightarrow \infty} \beta^T \frac{\partial U(x_T^*, x_{T+1}^*)}{\partial x} x_T^* = 0$$

which is exactly the transversality condition: a “boundary condition at infinity.”